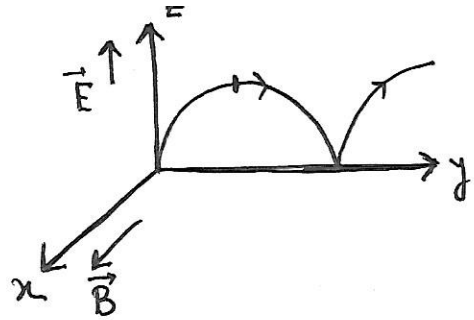
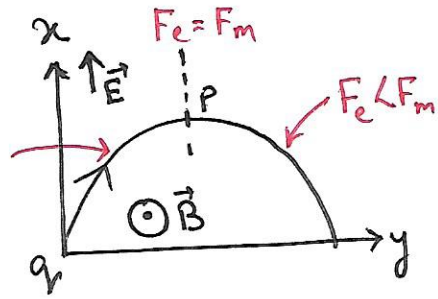
1. We have discussed the motion of charged particle (charge q and mass 'm') in the presence ($\vec{E} = E\hat{z}$ and $\vec{B} = B\hat{x}$, say)



The motion of the charged particle usually follows a cycloid due to the competition between electric and magnetic forces F_e and F_m. For example, F_e tries to force the charge move upward along z, velocity of charged particle increases, hence $\vec{F}_m = q(\vec{v}\times\vec{B})$ also increases which tries to force the particle move along a circular path.

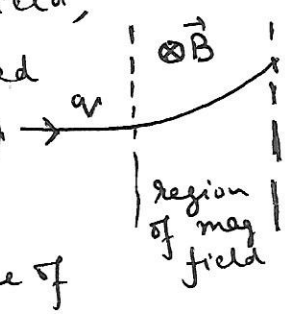


At peak point P, electric and magnetic forces are equal and beyond P, $F_e < F_m$.

[This is a preliminary discussion]

(a) Usually there is a zero deflection of motion of charged particle when $F_e = F_m$.

[usually in the presence of only magnetic field, there is a deflection in the motion of charged particle (as shown in adjacent figure) - it has been shown that a charged particle can trace out a circular motion in presence of $\vec{B}$ in some special cases]



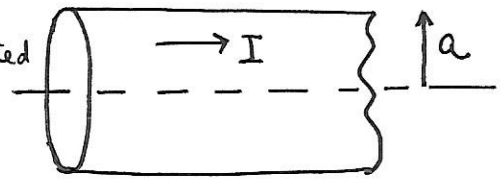
Anyways, for zero deflection, $qE = qvB \Rightarrow \boxed{v = \frac{E}{B}}$

(b) After the turns of electric field, the particle describes a circular path of radius R, the magnetic Lorentz force providing the centrifugal force of the circular motion (discussed in class)

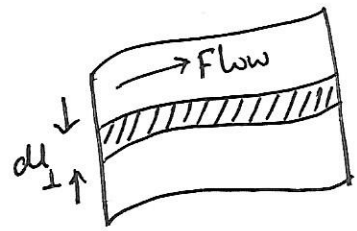
$$qvB = \frac{mv^2}{R} \Rightarrow \frac{q}{m} = \frac{v}{BR} = \frac{E}{B^2R} \quad \left[\because v = \frac{E}{B}\right]$$

2.

(a) Current is uniformly distributed over the surface.



The surface current density is defined as $\vec{K} = \dfrac{d\vec{I}}{dl_\perp}$



where $dl_\perp$ is the length perpendicular to the flow of current

For a cylindrical wire of circular crossection in which current I flows along the ~~direction~~ length of the cylinder, the total length perpendicular to the current flow is the circumference $2\pi a$

By definition, surface current of the wire is $K = \dfrac{I}{2\pi a}$

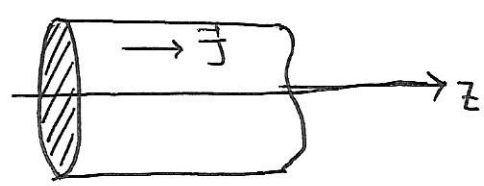
(b) It is given that $J \propto s$ or $J = k/s$ [s = distance from axis of cylinder, k = constant]

By definition, $I = \oint \vec{J} \cdot d\vec{a} = \oint J\, da_\perp$ [$da_\perp$ = magnitude of area perpendicular to the flow of current]

In cylindrical co-ordinates (s, φ, z)

$$dl_s = ds, \quad dl_\varphi = s\,d\varphi, \quad dl_z = dz$$

are infinitesimal lengths



We have to consider $d\vec{a} = dl_s\, dl_\varphi (\hat{z})$

$$= s\,ds\,d\varphi(\hat{z})$$

$da = s\,ds\,d\varphi$ is the shaded area

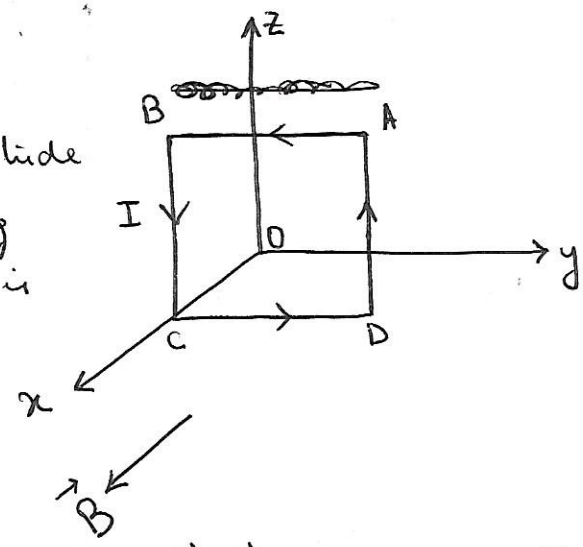
$$I = \int_0^a \frac{k}{s}\, s\,ds \int_0^{2\pi} d\varphi = 2\pi k a \Rightarrow k = \frac{I}{2\pi a}$$

$$\therefore \boxed{J = \frac{k}{s} = \frac{I}{2\pi a s}}$$

3.

$$\vec{B} = kz\,\hat{x}$$

A current of constant magnitude I flows anticlockwise along the square centered at O. Magnetic field is along x-direction



To calculate force

use $\vec{F} = \int (\vec{I} \times \vec{B})\, dl = I \int (d\vec{l} \times \vec{B})$ [I = const]

Then direction of force is given by the direction $d\vec{l} \times \vec{B}$

$$\vec{F}_{AB} = I\,[a(-\hat{y}) \times kz(\hat{x})] = Iakz\,\hat{z} = Ika \cdot \frac{a}{2}\,\hat{z} = \frac{Ika^2}{2}\,\hat{z}$$

[$\because z = \frac{a}{2}$ for AB]

$$\vec{F}_{CD} = I\,[a\hat{y} \times kz\,\hat{x}] = Iakz(-\hat{z}).$$ But $z = -\frac{a}{2}$ for CD

$$= Iak\left(-\frac{a}{2}\right) \times (-\hat{z})$$

$$= Ik\frac{a^2}{2}(\hat{z})$$

Now in similar manner $\vec{F}_{BC}$ and $\vec{F}_{DA}$ have same magnitudes but have opposite directions [$Iakz(-\hat{y})$ and $Iakz(\hat{y})$, respectively]

The net force along the upward direction is $\left(Ik\frac{a^2}{2} + Ik\frac{a^2}{2}\right)\hat{z}$

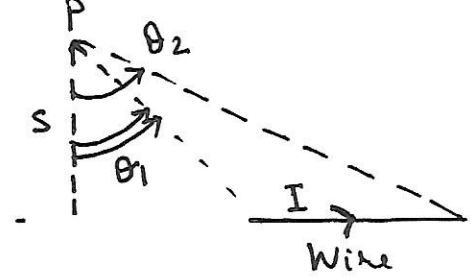
$$\boxed{\vec{F}_{net} = Ika^2\,\hat{z}}$$

4. We know that magnetic field due to a current carrying straight wire is given by Biot-Savart law as: ①

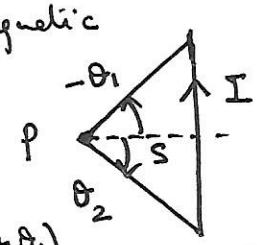
$$B = \frac{\mu_0 I}{4\pi s}(\sin\theta_2 - \sin\theta_1)$$ where θ_1 and θ_2 are angles subtended at the field point (say P) by the finite ends of the wire.

(done in class)

Note that in the expression above field point has been chosen such that θ_1 and θ_2 are both positive (same direction).



There can be field points (for e.g in the given problem) which subtend a +ve and a -ve value of θ, magnetic field at a distance s from the wire according to Biot Savart law would read

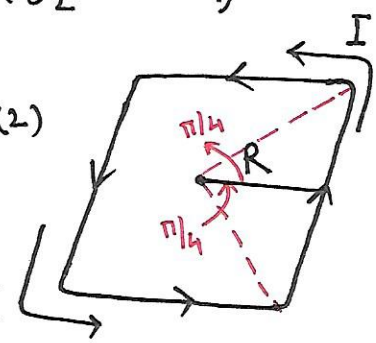


$$B = \frac{\mu_0 I}{4\pi s}\int_{-\theta_1}^{\theta_2} \cos\theta \, d\theta = \frac{\mu_0 I}{4\pi s}(\sin\theta_2 + \sin\theta_1) \qquad (2)$$

In the above case, since θ_1 and θ_2 were both +ve

$$B = \frac{\mu_0 I}{4\pi s}\int_{\theta_1}^{\theta_2} \cos\theta \, d\theta = \frac{\mu_0 I}{4\pi s}(\sin\theta_2 - \sin\theta_1)$$

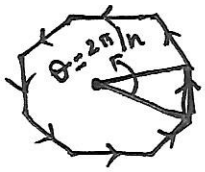
(a) In the given problem, using Eqn(2)



$$\left.\begin{aligned} B &= \frac{\mu_0 I}{4\pi R}\left(\sin\frac{\pi}{4} + \sin\frac{\pi}{4}\right) \\ \vec{B} &= \frac{\mu_0 I}{4\pi R}\left(\frac{2}{\sqrt{2}}\right)\hat{\varphi} \end{aligned}\right\} \text{at center of loop}$$

Clearly for all 4 sides, $\vec{B}_{center} = \left(\frac{\mu_0 I}{4\pi R}\right)\frac{2}{\sqrt{2}} \times 4 \, \hat{\varphi} = \boxed{\sqrt{2}\,\frac{\mu_0 I}{\pi R}\,\hat{\varphi}}$

(b) Now instead of a square, I have a regular n-sided polygon – angle subtended by any side of the polygon at the center is $\frac{2\pi}{n}$. Put $\theta_1 = \theta_2 = \frac{\pi}{n}$ in Eqn (2)



$$B = \frac{\mu_0 I n}{4\pi R}\left[\sin\left(\frac{\pi}{n}\right) + \sin\left(\frac{\pi}{n}\right)\right] = \left(\frac{\mu_0 I n}{4\pi R}\right) 2\sin\left(\frac{\pi}{n}\right) = \frac{\mu_0 I}{2\pi R}\sin\left(\frac{\pi}{n}\right) \times n$$

(n = no. of sides of polygon)

(c) For $n \to \infty$, $\sin\left(\frac{\pi}{n}\right) \to \frac{\pi}{n}$

$$B = \frac{\mu_0 n I}{2\pi R} \frac{\pi}{n} = \frac{\mu_0 I}{2R} \quad \text{——— (3)}$$

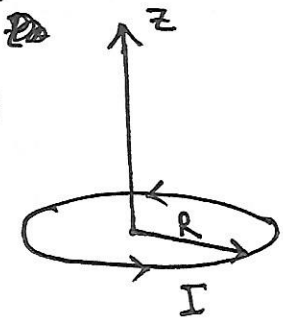
Now this field (3) is the same as the field at the center of a circular current-carrying loop.

It was shown in class that due to a circular loop, $\vec{B}$ is given by

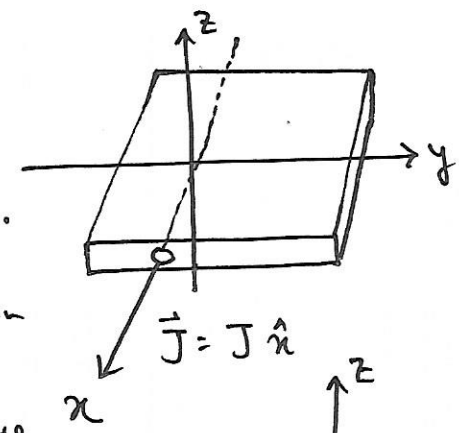
$$B = \frac{\mu_0}{4\pi} \frac{2\pi I R^2}{(R^2+z^2)^{3/2}}$$

[R: radius of the loop

at center $R=0$, $B = \frac{\mu_0 I}{2R}$ which is same as (3)



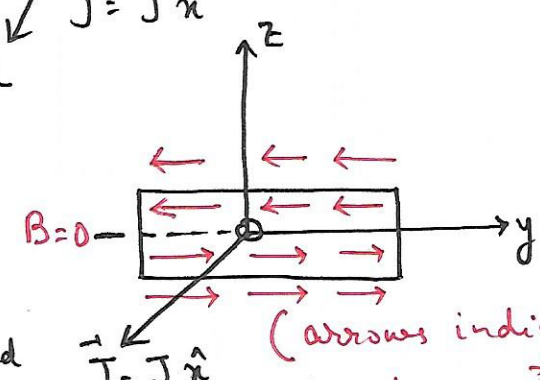
5. Use Biot Savart law to determine the direction of the magnetic field $\vec{B}$ due to current density $\vec{J}$ and at point above and below $z=0$ plane.



$$\vec{B} = \frac{\mu_0}{4\pi} \int \frac{\vec{J} \times \hat{r}}{r^2} d\tau'$$

[$\hat{r}$: separation vector from volume current source and field point]

Clearly $\vec{J} \times \hat{r}$ points along $(-\hat{y})$ for points above $z=0$ and along $(-\hat{y})$ for points below $z=0$. This can be verified by simple right hand rule.

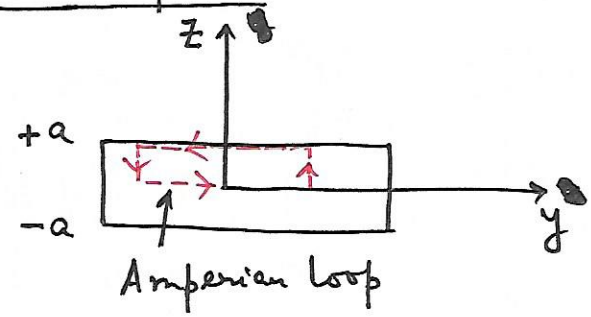


(arrows indicate direction of $\vec{B}$ due to $\vec{J}$ in an infinite slab of finite thickness)

Now that we have determined the direction of $\vec{B}$, we would find the magnitude of $\vec{B}$ by <u>Ampere's circuital law</u>.

Now by symmetry $\vec{B} = \vec{0}$ in $z = 0$ plane.

<u>Inside the slab</u> $0 < z < a$



Consider an amperian loop of length 'l' as shown inside the slab.

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc} \Rightarrow B \cdot l = \mu_0 J \times z \times l \quad [B = 0 \text{ at } z = 0]$$

(Ampere's law) $\Rightarrow B = \mu_0 J z$ (magnitude of $\vec{B}$)

Considering directions from Biot-Savart law, we can write

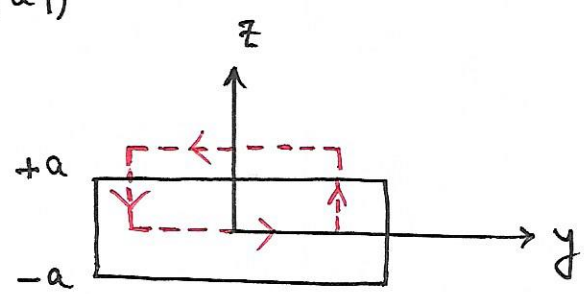
$$\boxed{\vec{B}_{in} = -\mu_0 J z (\hat{y})} \quad (-a < z < a)$$

$$\cancel{= -\mu_0 J a (\hat{y}) \quad [z > 0]}$$

$$\cancel{= +\mu_0 J a (\hat{y}) \quad [z < 0]}$$

<u>Outside the slab</u> $(z > |a|)$

Consider again amperian loop of length l.



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$$

$$Bl = \mu_0 J \times a \times l \quad [I_{enc} \text{ this time is } J \times a \times l]$$

$$B = \mu_0 J a$$

$$\therefore \boxed{\vec{B}_{out} = -\mu_0 J a (\hat{y})}$$

$$\vec{B}_{out} = -\mu_0 J a \hat{y} \quad z > a$$

$$= +\mu_0 J a \hat{y} \quad z > -a$$